

APPLICATION OF DERIVATIVES

1.	If $f(x) = a(x - \cos x)$ is strictly decreasing in $\mathbb{R}$, then 'a' belongs to (a) $\{0\}$ (b) $(0, \infty)$ (c) $(-\infty, 0)$ (d) $(-\infty, \infty)$
2.	Show that the function $f(x) = \frac{16 \sin x}{4 + \cos x} - x$, is strictly decreasing in $\left(\frac{\pi}{2}, \pi\right)$.
3.	If the circumference of circle is increasing at the constant rate, prove that rate of change of area of circle is directly proportional to its radius.
4.	Find the points on the curve $6y = x^3 + 2$ at which ordinate is changing 8 times as fast as abscissa.
5.	Find the maximum and minimum values of the function given by $f(x) = 5 + \sin 2x$.
6.	The interval in which the function $f(x) = 2x^3 + 9x^2 + 12x - 1$ is decreasing, is (a) $(-1, \infty)$ (b) $(-2, -1)$ (c) $(-\infty, -2)$ (d) $[-1, 1]$
7.	The function $f(x) = x x$, $x \in \mathbb{R}$ is differentiable (a) only at $x = 0$ (b) only at $x = 1$ (c) in $\mathbb{R}$ (d) in $\mathbb{R} - \{0\}$
8.	The median of an equilateral triangle is increasing at the rate of $2\sqrt{3}$ cm/s. Find the rate at which its side is increasing.
9.	Sum of two numbers is 5. If the sum of the cubes of these numbers is least, then find the sum of the squares of these numbers.
10.	For the function $f(x) = x^3$, $x = 0$ is a point of: (A) local maxima (B) local minima (C) non-differentiability (D) inflexion

11.	If the radius of a circle is increasing at the rate of 0.5 cm/s, then the rate of increase of its circumference is : (A) $\frac{2\pi}{3}$ cm/s (B) π cm/s (C) $\frac{4\pi}{3}$ cm/s (D) 2π cm/s
12.	Given that $f(x) = \frac{\log x}{x}$, find the point of local maximum of $f(x)$.
13.	Find the intervals in which the function $f(x) = x^4 - 4x^3 + 4x^2 + 15$, is strictly increasing.
14.	Show that the function $f(x) = 4x^3 - 18x^2 + 27x - 7$ has neither maxima nor minima.
15.	The function $f(x) = x^3 - 3x^2 + 12x - 18$ is : (A) strictly decreasing on $\mathbb{R}$ (B) strictly increasing on $\mathbb{R}$ (C) neither strictly increasing nor strictly decreasing on $\mathbb{R}$ (D) strictly decreasing on $(-\infty, 0)$
16.	If M and m denote the local maximum and local minimum values of the function $f(x) = x + \frac{1}{x}$ ($x \neq 0$) respectively, find the value of $(M - m)$.
17.	Show that $f(x) = e^x - e^{-x} + x - \tan^{-1} x$ is strictly increasing in its domain.
18.	The rate of change of surface area of a sphere with respect to its radius 'r', when $r = 4$ cm, is : (A) 64π cm²/cm (B) 48π cm²/cm (C) 32π cm²/cm (D) 16π cm²/cm

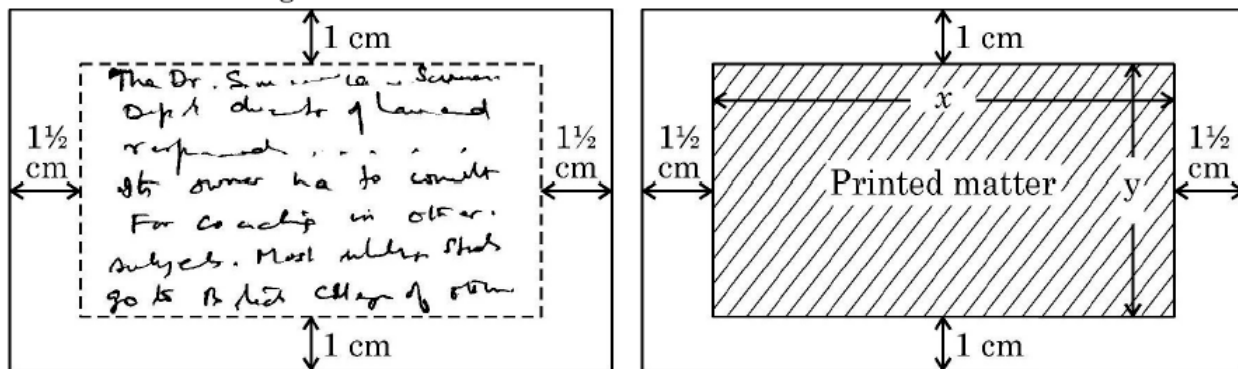
19.	Find the intervals in which the function $f(x) = \frac{\log x}{x}$ is strictly increasing or strictly decreasing.
20.	Find the absolute maximum and absolute minimum values of the function f given by $f(x) = \frac{x}{2} + \frac{2}{x}$, on the interval $[1, 2]$.
21.	It is given that function $f(x) = x^4 - 62x^2 + ax + 9$ attains local maximum value at $x = 1$. Find the value of 'a', hence obtain all other points where the given function $f(x)$ attains local maximum or local minimum values.
22.	The perimeter of a rectangular metallic sheet is 300 cm. It is rolled along one of its sides to form a cylinder. Find the dimensions of the rectangular sheet so that volume of cylinder so formed is maximum.
23.	The function $f(x) = kx - \sin x$ is strictly increasing for (A) $k > 1$ (B) $k < 1$ (C) $k > -1$ (D) $k < -1$
24.	Find the interval in which the function $f(x) = x^4 - 4x^3 + 10$ is strictly decreasing.
25.	The volume of a cube is increasing at the rate of $6 \text{ cm}^3/\text{s}$. How fast is the surface area of cube increasing, when the length of an edge is 8 cm ?
26.	The function $f(x) = \frac{x}{2} + \frac{2}{x}$ has a local minima at x equal to : (A) 2 (B) 1 (C) 0 (D) -2
27.	Given a curve $y = 7x - x^3$ and x increases at the rate of 2 units per second. The rate at which the slope of the curve is changing, when $x = 5$ is : (A) -60 units/sec (B) 60 units/sec (C) -70 units/sec (D) -140 units/sec

28.	The area of the circle is increasing at a uniform rate of $2 \text{ cm}^2/\text{sec}$. How fast is the circumference of the circle increasing when the radius $r = 5 \text{ cm}$?
29.	The absolute maximum value of function $f(x) = x^3 - 3x + 2$ in $[0, 2]$ is : (A) 0 (B) 2 (C) 4 (D) 5
30.	Find the intervals in which function $f(x) = 5x^{\frac{3}{2}} - 3x^{\frac{5}{2}}$ is (i) increasing (ii) decreasing.
31.	The side of an equilateral triangle is increasing at the rate of 3 cm/s . At what rate its area increasing when the side of the triangle is 15 cm ?
32.	Find the absolute maximum and absolute minimum of function $f(x) = 2x^3 - 15x^2 + 36x + 1$ on $[1, 5]$.
33.	The function $f(x) = x^2 - 4x + 6$ is increasing in the interval (A) $(0, 2)$ (B) $(-\infty, 2]$ (C) $[1, 2]$ (D) $[2, \infty)$
34.	The values of λ so that $f(x) = \sin x - \cos x - \lambda x + C$ decreases for all real values of x are : (A) $1 < \lambda < \sqrt{2}$ (B) $\lambda \geq 1$ (C) $\lambda \geq \sqrt{2}$ (D) $\lambda < 1$
35.	Find the values of 'a' for which $f(x) = \sin x - ax + b$ is increasing on $\mathbb{R}$.
36.	If $f(x) = 2x + \cos x$, then $f(x)$: (A) has a maxima at $x = \pi$ (B) has a minima at $x = \pi$ (C) is an increasing function (D) is a decreasing function
37.	Find the least value of 'a' so that $f(x) = 2x^2 - ax + 3$ is an increasing function on $[2, 4]$.

38.	If $f(x) = x + \frac{1}{x}$, $x \geq 1$, show that f is an increasing function.
39.	A cylindrical water container has developed a leak at the bottom. The water is leaking at the rate of $5 \text{ cm}^3/\text{s}$ from the leak. If the radius of the container is 15 cm, find the rate at which the height of water is decreasing inside the container, when the height of water is 2 metres.
40.	The relation between the height of the plant (y cm) with respect to exposure to sunlight is governed by the equation $y = 4x - \frac{1}{2}x^2$, where x is the number of days exposed to sunlight. (i) Find the rate of growth of the plant with respect to sunlight. 2 (ii) In how many days will the plant attain its maximum height ? What is the maximum height ? 3
41.	The slope of the curve $y = -x^3 + 3x^2 + 8x - 20$ is maximum at : (A) (1, -10) (B) (1, 10) (C) (10, 1) (D) (-10, 1)
42.	Surface area of a balloon (spherical), when air is blown into it, increases at a rate of $5 \text{ mm}^2/\text{s}$. When the radius of the balloon is 8 mm, find the rate at which the volume of the balloon is increasing.
43.	Find the value of 'a' for which $f(x) = \sqrt{3} \sin x - \cos x - 2ax + 6$ is decreasing in $\mathbb{R}$.
44.	Let $f(x) = x^2$, $x \in \mathbb{R}$. Then, which of the following statements is incorrect ? (A) Minimum value of f does not exist. (B) There is no point of maximum value of f in $\mathbb{R}$. (C) f is continuous at $x = 0$. (D) f is differentiable at $x = 0$.
45.	Determine those values of x for which $f(x) = \frac{2}{x} - 5$, $x \neq 0$ is increasing or decreasing.
CASE STUDY	

46.

A rectangular visiting card is to contain 24 sq.cm. of printed matter. The margins at the top and bottom of the card are to be 1 cm and the margins on the left and right are to be $1\frac{1}{2}$ cm as shown below :



On the basis of the above information, answer the following questions :

- (i) Write the expression for the area of the visiting card in terms of x .
- (ii) Obtain the dimensions of the card of minimum area.

47.

A carpenter needs to make a wooden cuboidal box, closed from all sides, which has a square base and fixed volume. Since he is short of the paint required to paint the box on completion, he wants the surface area to be minimum.

On the basis of the above information, answer the following questions :

- (i) Taking length = breadth = x m and height = y m, express the surface area (S) of the box in terms of x and its volume (V), which is constant. 1
- (ii) Find $\frac{dS}{dx}$. 1
- (iii) (a) Find a relation between x and y such that the surface area (S) is minimum. 2

OR

- (iii) (b) If surface area (S) is constant, the volume (V) = $\frac{1}{4}(Sx - 2x^3)$, x being the edge of base. Show that volume (V) is maximum for $x = \sqrt{\frac{S}{6}}$. 2

48.

A store has been selling calculators at ₹ 350 each. A market survey indicates that a reduction in price (p) of calculator increases the number of units (x) sold. The relation between the price and quantity sold is given by the demand function $p = 450 - \frac{1}{2}x$.

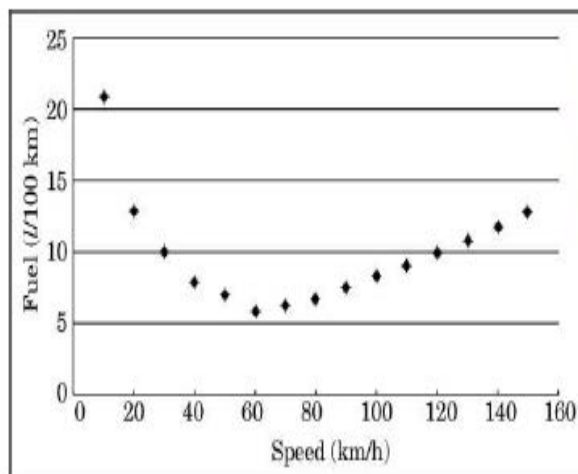


Based on the above information, answer the following questions :

- Determine the number of units (x) that should be sold to maximise the revenue $R(x) = xp(x)$. Also, verify the result.
- What rebate in price of calculator should the store give to maximise the revenue ?

49.

Overspeeding increases fuel consumption and decreases fuel economy as a result of tyre rolling friction and air resistance. While vehicles reach optimal fuel economy at different speeds, fuel mileage usually decreases rapidly at speeds above 80 km/h.



The relation between fuel consumption F (l/100 km) and speed V (km/h) under some constraints is given as $F = \frac{V^2}{500} - \frac{V}{4} + 14$.

On the basis of the above information, answer the following questions :

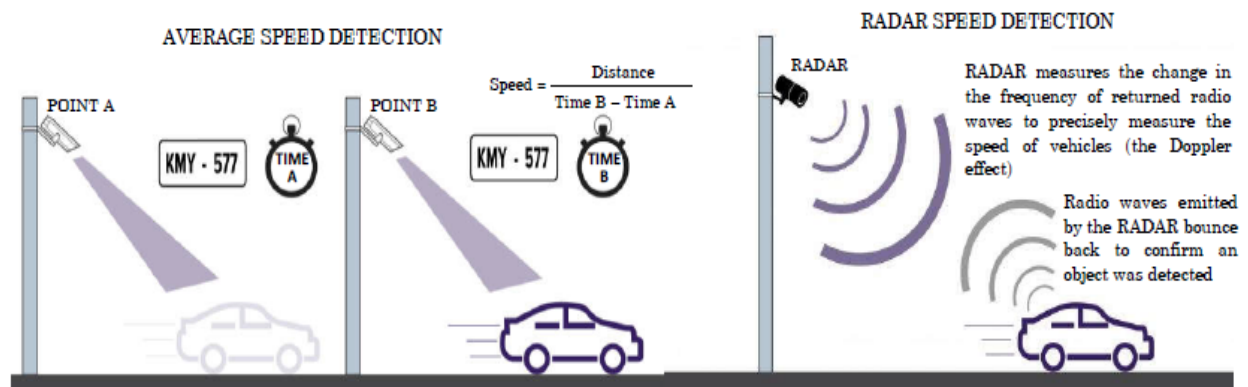
- (i) Find F , when $V = 40$ km/h.
- (ii) Find $\frac{dF}{dV}$.
- (iii) (a) Find the speed V for which fuel consumption F is minimum.

OR

- (iii) (b) Find the quantity of fuel required to travel 600 km at the speed V at which $\frac{dF}{dV} = -0.01$.

50.

The traffic police has installed Over Speed Violation Detection (OSVD) system at various locations in a city. These cameras can capture a speeding vehicle from a distance of 300 m and even function in the dark.



A camera is installed on a pole at the height of 5 m. It detects a car travelling away from the pole at the speed of 20 m/s. At any point, x m away from the base of the pole, the angle of elevation of the speed camera from the car C is θ .

On the basis of the above information, answer the following questions :

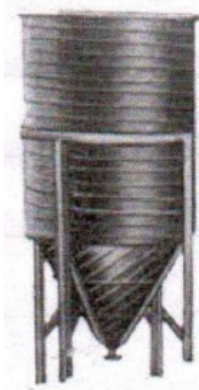
- (i) Express θ in terms of height of the camera installed on the pole and x .
- (ii) Find $\frac{d\theta}{dx}$.
- (iii) (a) Find the rate of change of angle of elevation with respect to time at an instant when the car is 50 m away from the pole.

OR

- (iii) (b) If the rate of change of angle of elevation with respect to time of another car at a distance of 50 m from the base of the pole is $\frac{3}{101}$ rad/s, then find the speed of the car.

51.

A tank, as shown in the figure below, formed using a combination of a cylinder and a cone, offers better drainage as compared to a flat bottomed tank.



A tap is connected to such a tank whose conical part is full of water. Water is dripping out from a tap at the bottom at the uniform rate of $2 \text{ cm}^3/\text{s}$. The semi-vertical angle of the conical tank is 45° .

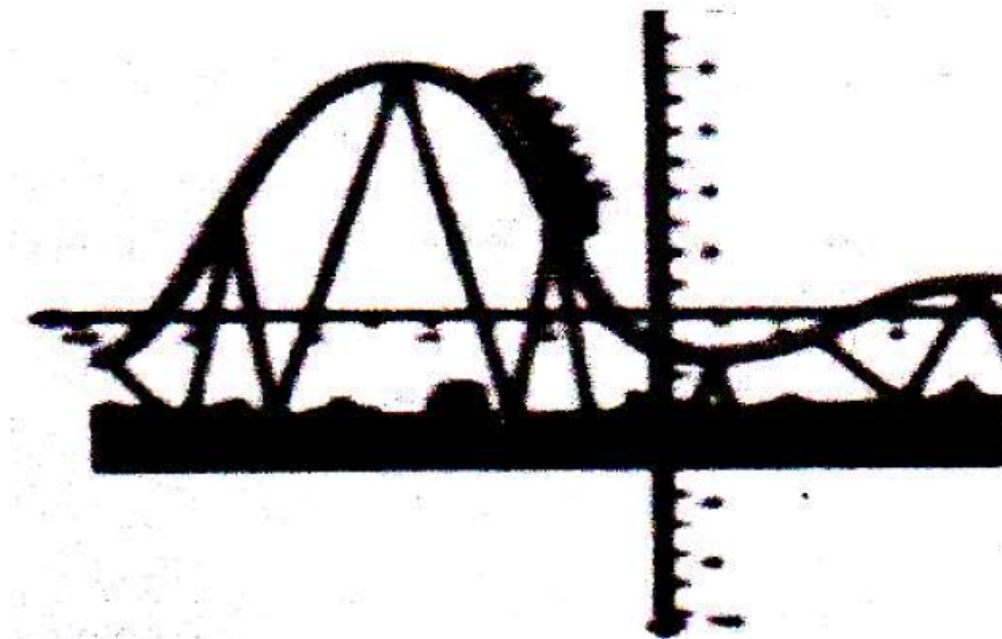
On the basis of given information, answer the following questions :

- (i) Find the volume of water in the tank in terms of its radius r .
- (ii) Find rate of change of radius at an instant when $r = 2\sqrt{2} \text{ cm}$.
- (iii) (a) Find the rate at which the wet surface of the conical tank is decreasing at an instant when radius $r = 2\sqrt{2} \text{ cm}$.

OR

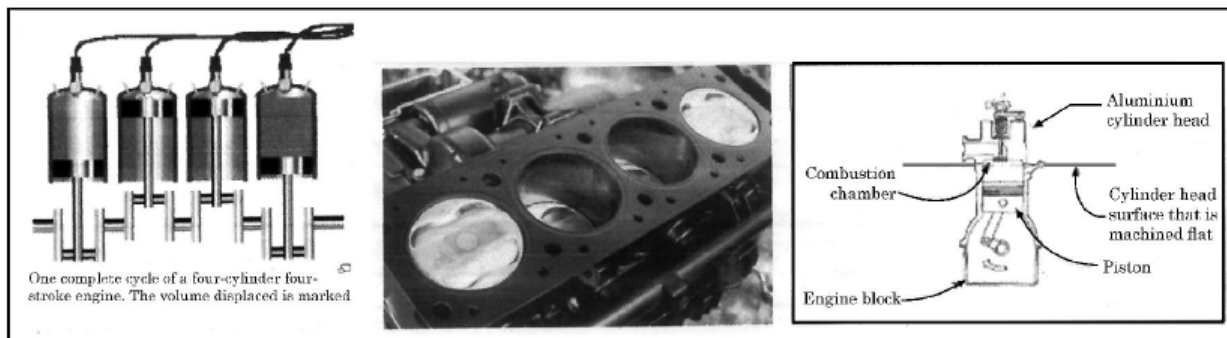
- (iii) (b) Find the rate of change of height 'h' at an instant when slant height is 4 cm.

The equation of the path traced by a roller-coaster is given by the polynomial $f(x) = a(x + 9)(x + 1)(x - 3)$. If the roller-coaster crosses y-axis at a point $(0, -1)$, answer the following :



- (i) Find the value of 'a'.
- (ii) Find $f''(x)$ at $x = 1$.

- 52.** Engine displacement is the measure of the cylinder volume swept by all the pistons of a piston engine. The piston moves inside the cylinder bore



The cylinder bore in the form of circular cylinder open at the top is to be made from a metal sheet of area $75\pi \text{ cm}^2$.

Based on the above information, answer the following questions :

- (i) If the radius of cylinder is $r \text{ cm}$ and height is $h \text{ cm}$, then write the volume V of cylinder in terms of radius r .
- (ii) Find $\frac{dV}{dr}$.
- (iii) (a) Find the radius of cylinder when its volume is maximum.

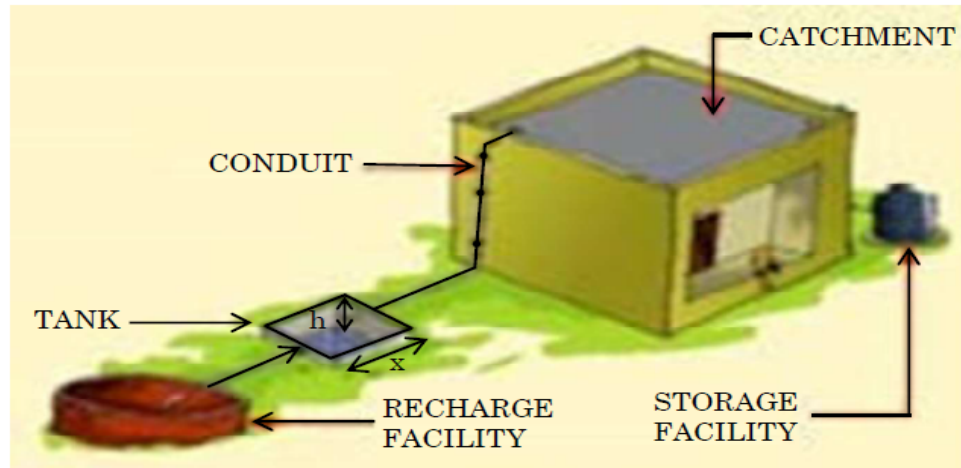
OR

- (b) For maximum volume, $h > r$. State true or false and justify.

53.

In order to set up a rain water harvesting system, a tank to collect rain water is to be dug. The tank should have a square base and a capacity of 250 m^3 . The cost of land is ₹ 5,000 per square metre and cost of digging increases with depth and for the whole tank, it is ₹ $40,000 h^2$, where h is the depth of the tank in metres. x is the side of the square base of the tank in metres.

ELEMENTS OF A TYPICAL RAIN WATER HARVESTING SYSTEM



- (iii) (a) Find the value of x for which cost C is minimum.

OR

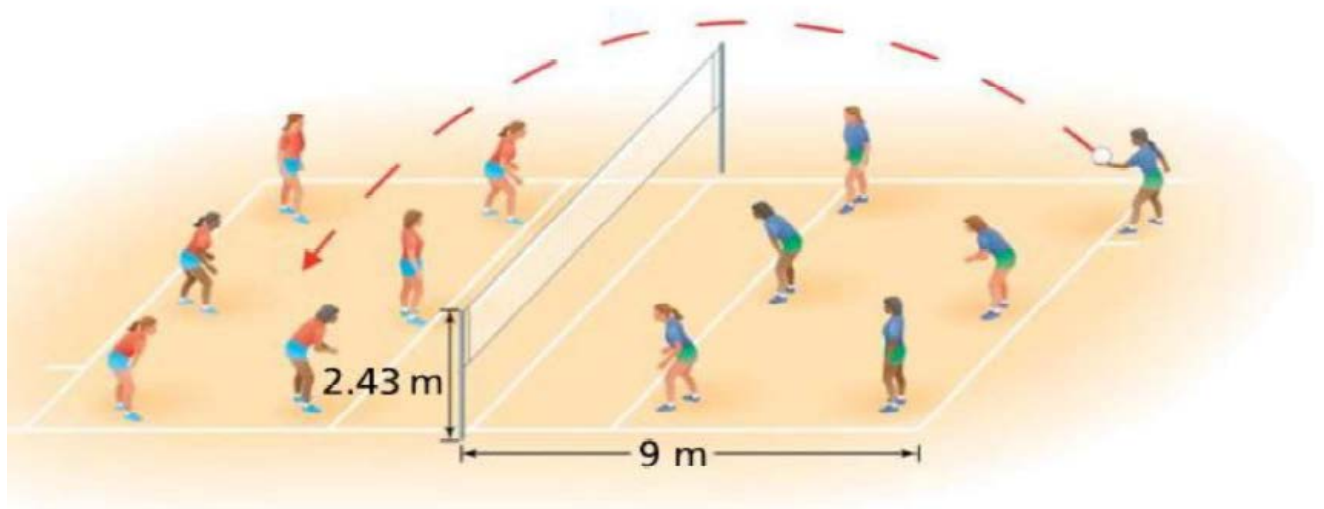
- (iii) (b) Check whether the cost function $C(x)$ expressed in terms of x is increasing or not, where $x > 0$.

Based on the above information, answer the following questions :

- (i) Find the total cost C of digging the tank in terms of x .
- (ii) Find $\frac{dC}{dx}$.

54.

A volleyball player serves the ball which takes a parabolic path given by the equation $h(t) = -\frac{7}{2}t^2 + \frac{13}{2}t + 1$, where $h(t)$ is the height of ball at any time t (in seconds), ($t \geq 0$).

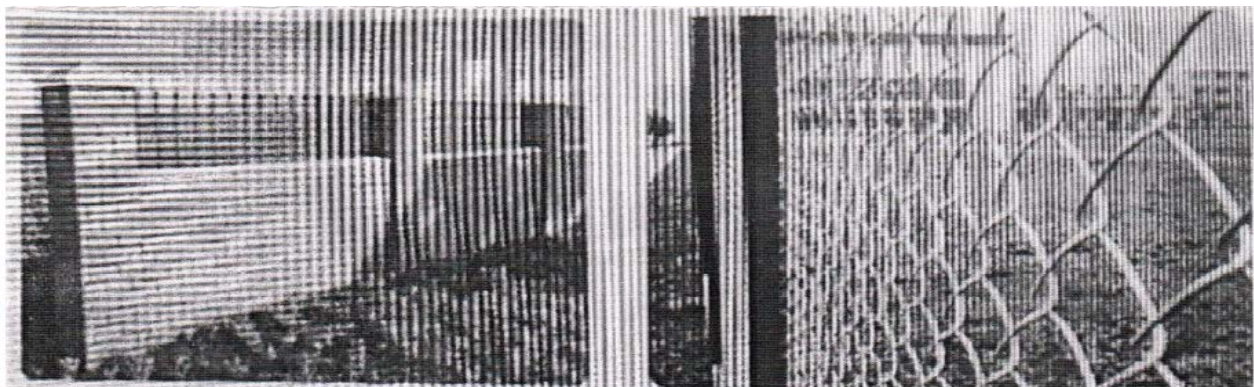


Based on the above information, answer the following questions :

- (i) Is $h(t)$ a continuous function ? Justify.
- (ii) Find the time at which the height of the ball is maximum.

55.

Sooraj's father wants to construct a rectangular garden using a brick wall on one side of the garden and wire fencing for the other three sides as shown in the figure. He has 200 metres of fencing wire.

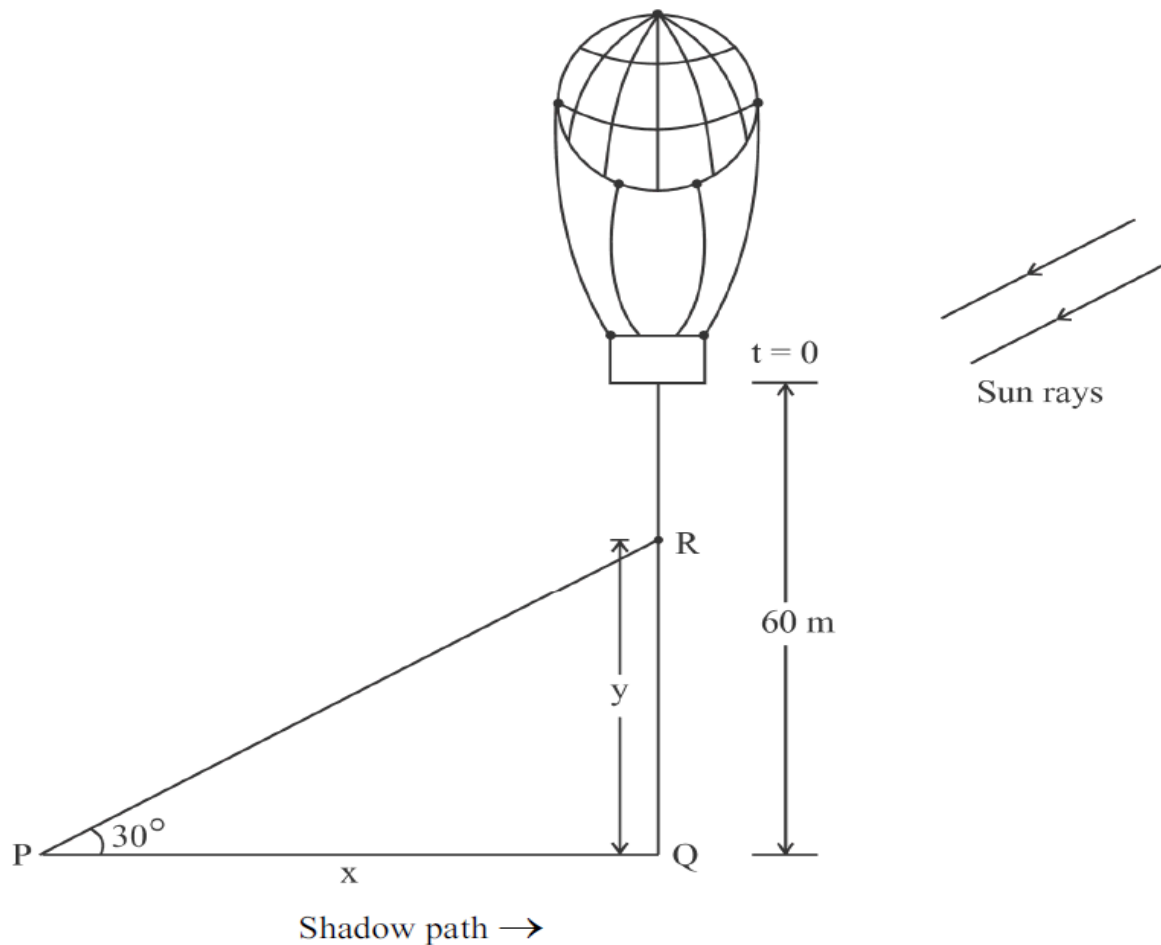


Based on the above information, answer the following questions :

- (i) Let ' x ' metres denote the length of the side of the garden perpendicular to the brick wall and ' y ' metres denote the length of the side parallel to the brick wall. Determine the relation representing the total length of fencing wire and also write $A(x)$, the area of the garden.
- (ii) Determine the maximum value of $A(x)$.

56.

A sandbag is dropped from a balloon at a height of 60 metres.



When the angle of elevation of the sun is 30° , the position of the sandbag is given by the equation $y = 60 - 4.9 t^2$, where y is the height of the sandbag above the ground and t is the time in seconds.

On the basis of the above information, answer the following questions :

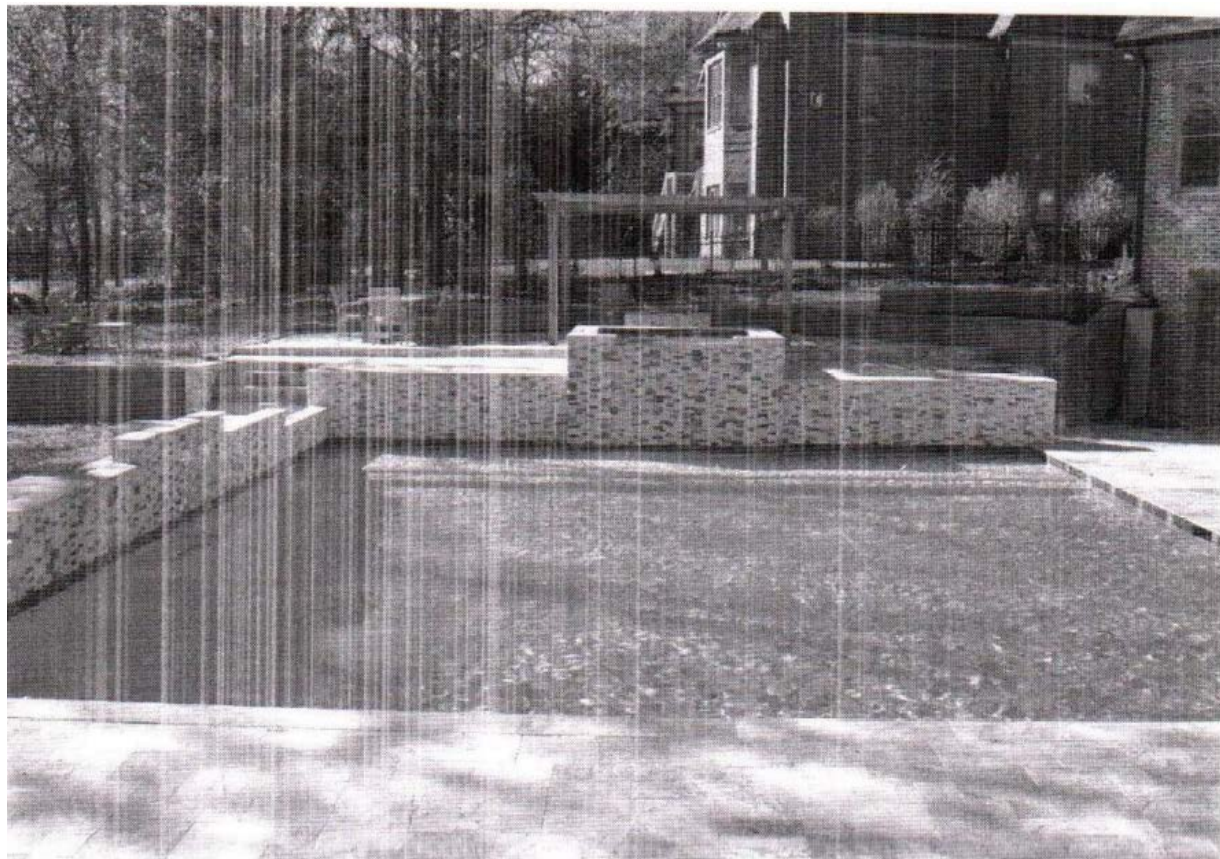
- (i) Find the relation between x and y , where x is the distance of the shadow at P from the point Q and y is the height of the sandbag above the ground.
- (ii) After how much time will the sandbag be 35 metres above the ground ?
- (iii) (a) Find the rate at which the shadow of the sandbag is travelling along the ground when the sandbag is at a height of 35 metres.

OR

- (iii) (b) How fast is the height of the sandbag decreasing when 2 seconds have elapsed ?

57.

A housing society wants to commission a swimming pool for its residents. For this, they have to purchase a square piece of land and dig this to such a depth that its capacity is 250 cubic metres. Cost of land is ₹ 500 per square metre. The cost of digging increases with the depth and cost for the whole pool is ₹ $4000(\text{depth})^2$.



Suppose the side of the square plot is x metres and depth is h metres.

On the basis of the above information, answer the following questions :

- (i) Write cost $C(h)$ as a function in terms of h .
- (ii) Find critical point.
- (iii) (a) Use second derivative test to find the value of h for which cost of constructing the pool is minimum. What is the minimum cost of construction of the pool ?

OR

- (iii) (b) Use first derivative test to find the depth of the pool so that cost of construction is minimum. Also, find relation between x and h for minimum cost.

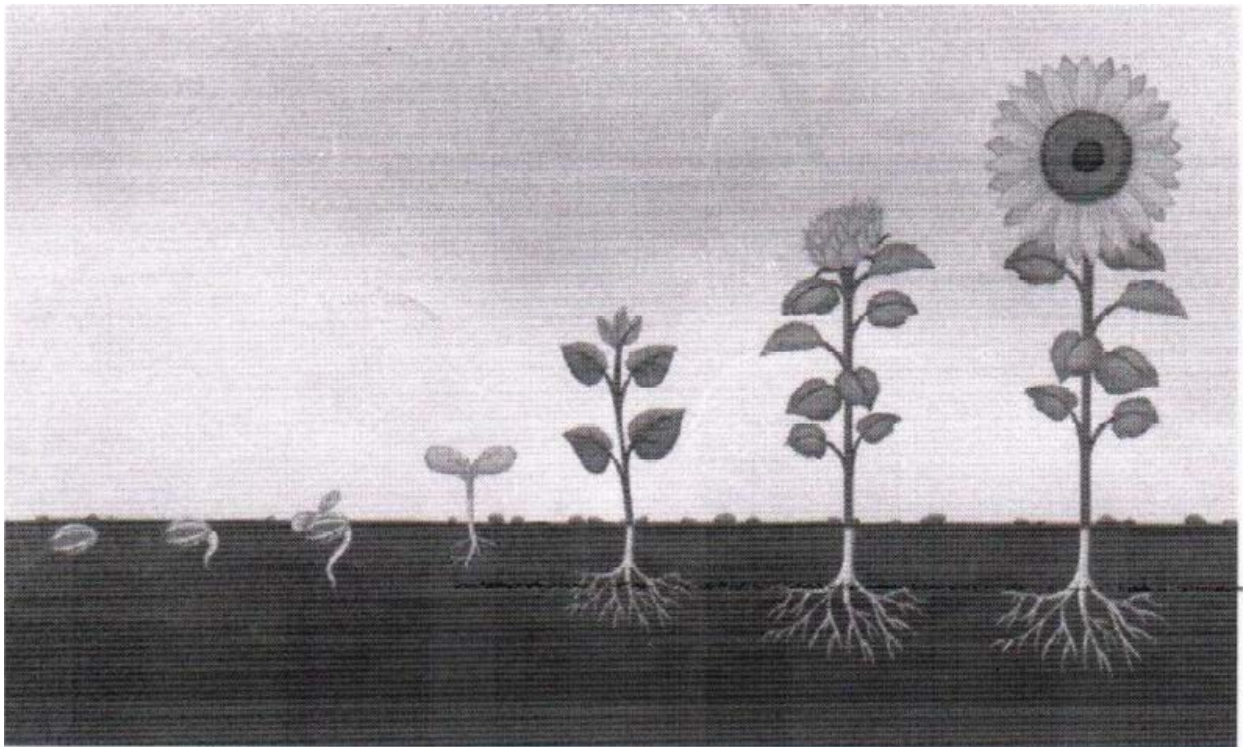
58.

In an agricultural institute, scientists do experiments with varieties of seeds to grow them in different environments to produce healthy plants and get more yield.

A scientist observed that a particular seed grew very fast after germination. He had recorded growth of plant since germination and he said that its growth can be defined by the function

$$f(x) = \frac{1}{3}x^3 - 4x^2 + 15x + 2, \quad 0 \leq x \leq 10$$

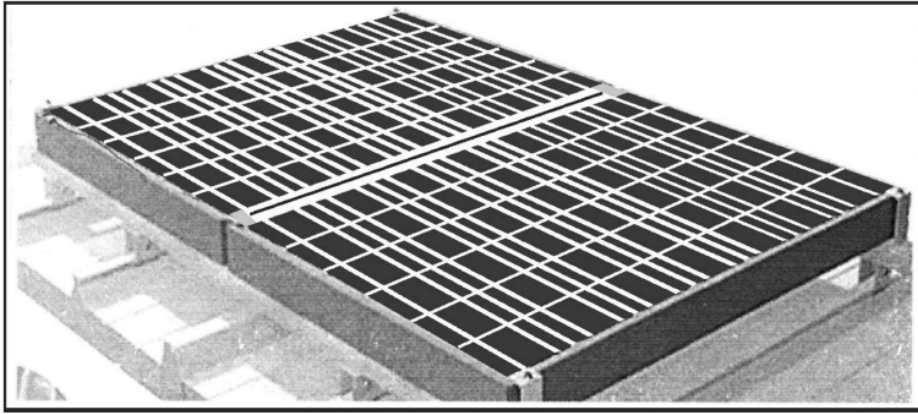
where x is the number of days the plant is exposed to sunlight.



On the basis of the above information, answer the following questions :

- (i) What are the critical points of the function $f(x)$?
- (ii) Using second derivative test, find the minimum value of the function.

59.



A technical company is designing a rectangular solar panel installation on a roof using 300 metres of boundary material. The design includes a partition running parallel to one of the sides dividing the area (roof) into two sections.

Let the length of the side perpendicular to the partition be x metres and with parallel to the partition be y metres.

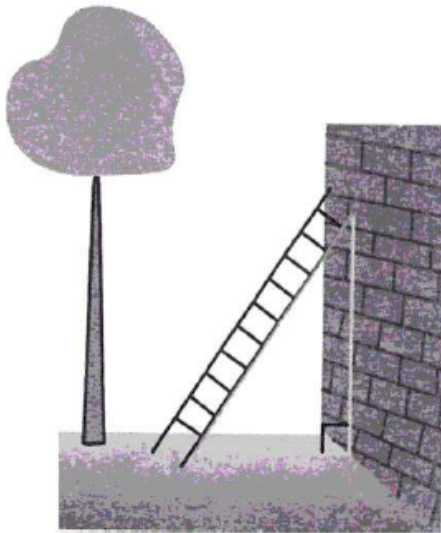
Based on this information, answer the following questions :

- (i) Write the equation for the total boundary material used in the boundary and parallel to the partition in terms of x and y . 1
- (ii) Write the area of the solar panel as a function of x . 1
- (iii) (a) Find the critical points of the area function. Use second derivative test to determine critical points at the maximum area. Also, find the maximum area. 2

OR

- (iii) (b) Using first derivative test, calculate the maximum area the company can enclose with the 300 metres of boundary material, considering the parallel partition. 2

60.



A ladder of fixed length ' h ' is to be placed along the wall such that it is free to move along the height of the wall.

Based upon the above information, answer the following questions :

- (i) Express the distance (y) between the wall and foot of the ladder in terms of ' h ' and height (x) on the wall at a certain instant. Also, write an expression in terms of h and x for the area (A) of the right triangle, as seen from the side by an observer. 1

- (ii) Find the derivative of the area (A) with respect to the height on the wall (x), and find its critical point. 1

- (iii) (a) Show that the area (A) of the right triangle is maximum at the critical point. 2

OR

- (iii) (b) If the foot of the ladder whose length is 5 m, is being pulled towards the wall such that the rate of decrease of distance (y) is 2 m/s, then at what rate is the height on the wall (x) increasing, when the foot of the ladder is 3 m away from the wall ? 2

61.

Edge of a variable cube increases at the rate of 5 cm/s. The rate at which the surface area of the cube increases when the edge is 2 cm long is :

- (A) $24 \text{ cm}^2/\text{s}$ (B) $120 \text{ cm}^2/\text{s}$
 (C) $12 \text{ cm}^2/\text{s}$ (D) $5 \text{ cm}^2/\text{s}$

62.

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined as $f(x) = 2x - \sin x$, then f is :

- (A) a decreasing function (B) an increasing function
(C) maximum at $x = \frac{\pi}{2}$ (D) maximum at $x = 0$

63.

A small town is analyzing the pattern of a new street light installation. The lights are set up in such a way that the intensity of light at any point x metres from the start of the street can be modelled by $f(x) = e^x \sin x$, where x is in metres.

Based on the above, answer the following :

- (i) Find the intervals on which the $f(x)$ is increasing or decreasing, $x \in [0, \pi]$. 2
(ii) Verify, whether each critical point when $x \in [0, \pi]$ is a point of local maximum or local minimum or a point of inflexion. 2

64.

Show that of all the rectangles with a fixed perimeter, the square has the greatest area.